

Stat 380 Mini Project 1

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1. Load Data

1.1 Libraries

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.3

```
library(ggplot2)
```

1.2 Import Data Files

Each file is in wide format with one row per country and one column per year.

```
sanitation <- read_csv("../data/predictors/at_least_basic_sanitation_overall_access_percent.csv")
fertility <- read_csv("../data/predictors/children_per_woman_total_fertility.csv")
co2 <- read_csv("../data/predictors/co2_pcap_cons.csv")
happiness <- read_csv("../data/predictors/hapiscore_whr.csv")
income <- read_csv("../data/predictors/ilevels4_wb.csv", col_types = cols(.default = "c"))
poverty <- read_csv("../data/predictors/number_of_people_in_poverty.csv")
population <- read_csv("../data/response/pop.csv")
```

1.3 Dropping Year Columns

The year columns from 2023 onwards are dropped first. Each dataset is then changed to long format so that all of them share the same geo, name and year keys, which avoids duplicated year column names when joining.

```

drop_and_pivot <- function(df, value_name) {
  df |>
    select(-any_of(as.character(2023:2100))) |>
    pivot_longer(
      cols = -c(geo, name),
      names_to = "year",
      values_to = value_name
    ) |>
    mutate(year = as.integer(year))
}

sanitation_long <- drop_and_pivot(sanitation, "sanitation_access_pct")
fertility_long <- drop_and_pivot(fertility, "children_per_woman")
co2_long <- drop_and_pivot(co2, "co2_per_capita")
happiness_long <- drop_and_pivot(happiness, "happiness_score")
income_long <- drop_and_pivot(income, "income_level")
poverty_long <- drop_and_pivot(poverty, "people_in_poverty")
population_long <- drop_and_pivot(population, "population")

```

1.4 Join into a Single Dataset

```

combined <- list(
  population_long, sanitation_long, fertility_long, co2_long,
  happiness_long, income_long, poverty_long
) |>
  reduce(full_join, by = c("geo", "name", "year")) |>
  arrange(geo, year)

glimpse(combined)

```

Rows: 43,485

Columns: 10

```

$ geo      <chr> "afg", "afg", "afg", "afg", "afg", "afg", "afg", ~
$ name     <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Af~
$ year     <int> 1800, 1801, 1802, 1803, 1804, 1805, 1806, 1807, ~
$ population <dbl> 3280000, 3280000, 3280000, 3280000, 3280000, 328~
$ sanitation_access_pct <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ children_per_woman <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, ~
$ co2_per_capita <dbl> 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, ~
$ happiness_score <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~

```

```
$ income_level      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ people_in_poverty <dbl> 3082964, 3082964, 3082964, 3082964, 3082964, 308~
```

```
range(combined$year)
```

```
[1] 1800 2022
```

```
n_distinct(combined$geo)
```

```
[1] 195
```

```
head(combined)
```

```
# A tibble: 6 x 10
  geo   name      year population sanitation_access_pct children_per_woman
  <chr> <chr>    <int>      <dbl>          <dbl>              <dbl>
1 afg  Afghanistan 1800      3280000          NA                7
2 afg  Afghanistan 1801      3280000          NA                7
3 afg  Afghanistan 1802      3280000          NA                7
4 afg  Afghanistan 1803      3280000          NA                7
5 afg  Afghanistan 1804      3280000          NA                7
6 afg  Afghanistan 1805      3280000          NA                7
# i 4 more variables: co2_per_capita <dbl>, happiness_score <dbl>,
#   income_level <chr>, people_in_poverty <dbl>
```

2. Exploratory Data Analysis

2.1 Total Population

population is quantitative and discrete, measured as the total number of people living in a country. Values are non-negative whole numbers because population represents a count of people. This will be our response variable.

```
population_long |>
  summarise(
    n          = n(),
    n_missing  = sum(is.na(population)),
    pct_missing = round(100 * mean(is.na(population)), 1),
    first_year = min(year[!is.na(population)]),
    last_year  = max(year[!is.na(population)])
  )
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
  <int>   <int>     <dbl>     <int>    <int>
1 43485     22      0.1      1800     2022
```

There are 22 missing population observations, representing 0.1% of the data. Because the population data cover many countries across a long time period, the small number of missing observations appears to be limited missing data rather than a major issue with the dataset.

```
population_long |>
  filter(!is.na(population), population < 0) |>
  arrange(population)
```

```
# A tibble: 0 x 4
# i 4 variables: geo <chr>, name <chr>, year <int>, population <dbl>
```

There are no negative population values, so no observations need to be removed based on this check. The population data are clean with respect to impossible negative values.

```
population_long |>
  filter(!is.na(population)) |>
  summarise(
    mean  = mean(population),
    median = median(population),
    sd    = sd(population),
    min   = min(population),
    q1    = quantile(population, 0.25),
    q3    = quantile(population, 0.75),
    max   = max(population)
  )
```

```
# A tibble: 1 x 7
      mean median      sd  min    q1    q3    max
  <dbl>  <dbl>   <dbl> <dbl>  <dbl>  <dbl>  <dbl>
1 13637388. 1760301 67079427.   651 292768. 6243734 1426437267
```

```
population_2022 <- population_long |>
  filter(!is.na(population), year == 2022)

population_2022 |>
```

```

summarise(
  year      = unique(year),
  n_countries = n(),
  mean      = mean(population),
  median    = median(population),
  sd        = sd(population),
  min       = min(population),
  q1        = quantile(population, 0.25),
  q3        = quantile(population, 0.75),
  max       = max(population)
)

```

```

# A tibble: 1 x 9
  year n_countries    mean  median      sd   min     q1     q3    max
<int>   <int>    <dbl>  <dbl>   <dbl> <dbl>  <dbl>  <dbl>  <dbl>
1  2022      194 41134428. 9096444. 149845684. 9992 2107954. 3.04e7 1.43e9

```

In 2022, the dataset contains 194 countries. The mean population is about 41.1 million, while the median is about 9.1 million, showing that the distribution is strongly right-skewed. The standard deviation is about 149.8 million, reflecting the large differences in population size between countries. The middle 50% of countries have populations between approximately 2.1 million and 30.4 million. The maximum population is approximately 1.43 billion, indicating that a small number of highly populated countries have a large influence on the mean.

```

population_long |>
  filter(!is.na(population)) |>
  group_by(year) |>
  summarise(mean_population = mean(population)) |>
  ggplot(aes(year, mean_population)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  labs(
    x = "Year",
    y = "Mean population",
    title = "Mean Population by Year"
  ) +
  theme_minimal()

```

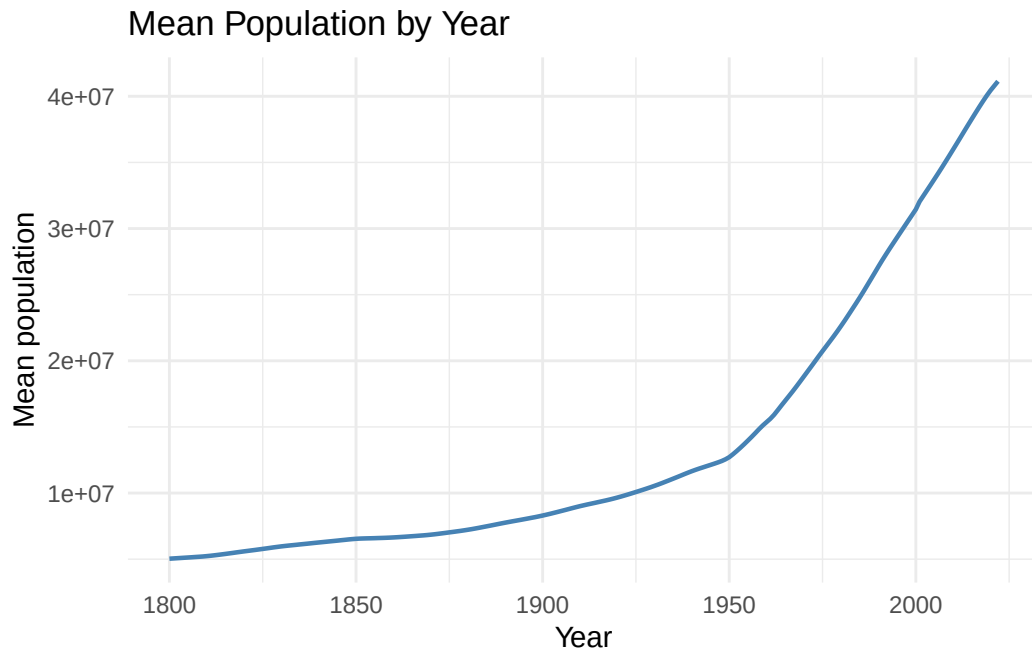


Figure 1: Mean population by year

The mean population increased throughout the period shown, with population growth becoming much more rapid after approximately 1950. This indicates substantial growth in the average population of countries over time, particularly during the second half of the twentieth century.

```
ggplot(population_2022, aes(population)) +  
  geom_histogram(bins = 25, fill = "blue", color = "white") +  
  labs(  
    x = "Population",  
    y = "Number of countries",  
    title = "Distribution of country populations, 2022"  
  ) +  
  theme_minimal()
```

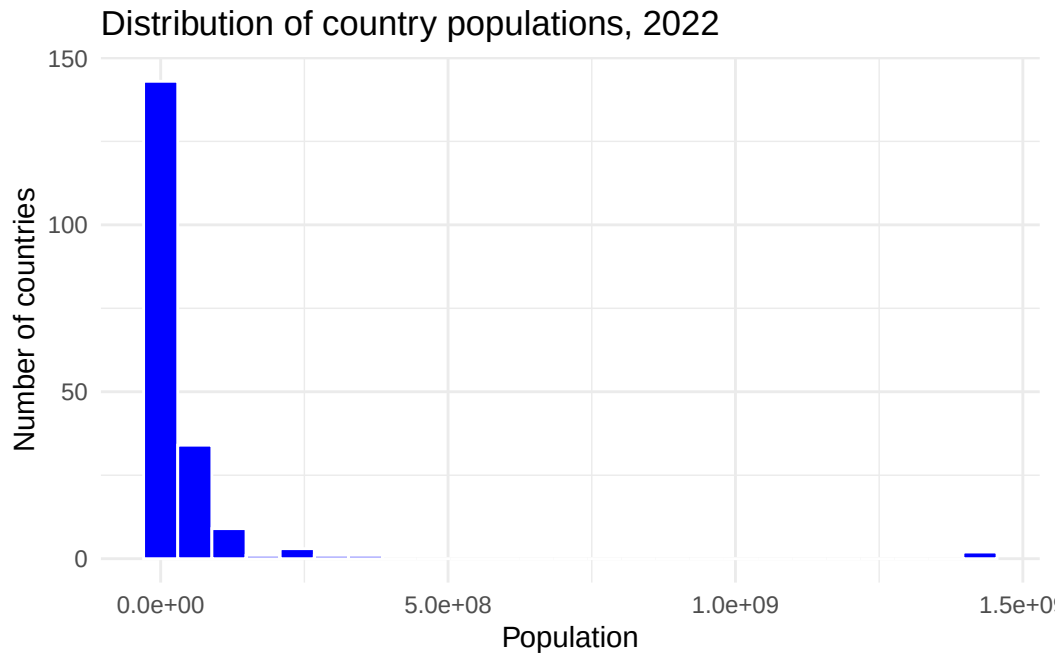


Figure 2: Distribution of country populations, 2022

The 2022 population distribution is strongly right-skewed. Most countries have relatively small populations, while only a few countries have populations much larger than the rest. The extreme value near 1.4 billion corresponds to one of the world's most populous countries and has a strong influence on the mean.

```
bind_rows(
  population_2022 |>
    slice_max(population, n = 5) |>
    mutate(group = "Highest"),
  population_2022 |>
    slice_min(population, n = 5) |>
    mutate(group = "Lowest")
) |>
select(group, name, population)
```

```
# A tibble: 10 x 3
  group  name      population
  <chr>  <chr>      <dbl>
1 Highest India    1425423212
2 Highest China    1425179569
3 Highest USA      341534046
```

4	Highest	Indonesia	278830529
5	Highest	Pakistan	243700667
6	Lowest	Tuvalu	9992
7	Lowest	Nauru	11801
8	Lowest	Palau	17759
9	Lowest	San Marino	34090
10	Lowest	Monaco	38931

Population is highly variable across countries and has increased substantially over time. The 2022 distribution is strongly right-skewed, with most countries having relatively small populations and a small number of countries having extremely large populations. India and China are clear high-end outliers, each having about 1.43 billion people, while several small island and European countries have populations below 40,000. Because of this strong skew and the influence of very large countries, the median is more representative of a typical country's population than the mean.

2.2 Children per Woman

`children_per_woman` is quantitative and continuous. It represents the average number of children a woman would have over her lifetime. Because it is an average, it can take decimal values rather than only whole numbers.

```
children_per_woman_long <- fertility_long

children_per_woman_long |>
  summarise(
    n = n(),
    n_missing = sum(is.na(children_per_woman)),
    pct_missing = round(100 * mean(is.na(children_per_woman)), 1),
    first_year = min(year[!is.na(children_per_woman)]),
    last_year = max(year[!is.na(children_per_woman)])
  )
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
<int>   <int>     <dbl>     <int>   <int>
1 43485         0         0       1800    2022
```

There are no missing observations for children per woman, so the variable has complete coverage across the dataset from 1800 to 2022.

```
children_per_woman_long |>
  filter(!is.na(children_per_woman), children_per_woman < 0) |>
  arrange(children_per_woman)
```

```
# A tibble: 0 x 4
# i 4 variables: geo <chr>, name <chr>, year <int>, children_per_woman <dbl>
```

There are no negative values for children per woman, so no observations need to be removed based on this check. The data are clean with respect to impossible negative values.

We also make sure the variable is stored numerically.

```
children_per_woman_long <- children_per_woman_long |>
  mutate(children_per_woman = as.numeric(children_per_woman))
```

```
children_per_woman_long |>
  filter(!is.na(children_per_woman)) |>
  summarise(
    mean    = mean(children_per_woman),
    median  = median(children_per_woman),
    sd      = sd(children_per_woman),
    min     = min(children_per_woman),
    q1      = quantile(children_per_woman, 0.25),
    q3      = quantile(children_per_woman, 0.75),
    max     = max(children_per_woman)
  )
```

```
# A tibble: 1 x 7
  mean median    sd  min    q1    q3  max
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  5.27  5.87  1.73  0.78  4.29  6.58  8.86
```

```
children_per_woman_2022 <- children_per_woman_long |>
  filter(!is.na(children_per_woman), year == 2022)

children_per_woman_2022 |>
  summarise(
    year      = unique(year),
    n_countries = n(),
    mean      = mean(children_per_woman),
```

```

median      = median(children_per_woman),
sd          = sd(children_per_woman),
min         = min(children_per_woman),
q1          = quantile(children_per_woman, 0.25),
q3          = quantile(children_per_woman, 0.75),
max         = max(children_per_woman)
)

```

A tibble: 1 x 9

	year	n_countries	mean	median	sd	min	q1	q3	max
	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022	195	2.50	2	1.25	0.78	1.54	3.28	6.26

In 2022, the mean number of children per woman was 2.50 and the median was 2.00. The standard deviation was 1.25, with values ranging from 0.78 to 6.26 children per woman. The middle 50% of countries had values between approximately 1.54 and 3.28. The mean being higher than the median indicates that the 2022 distribution is right-skewed, with a smaller number of countries having substantially higher fertility rates.

```

children_per_woman_long |>
  filter(!is.na(children_per_woman)) |>
  group_by(year) |>
  summarise(mean_children = mean(children_per_woman)) |>
  ggplot(aes(year, mean_children)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  labs(
    x = "Year",
    y = "Mean children per woman",
    title = "Mean children per woman by year"
  ) +
  theme_minimal()

```

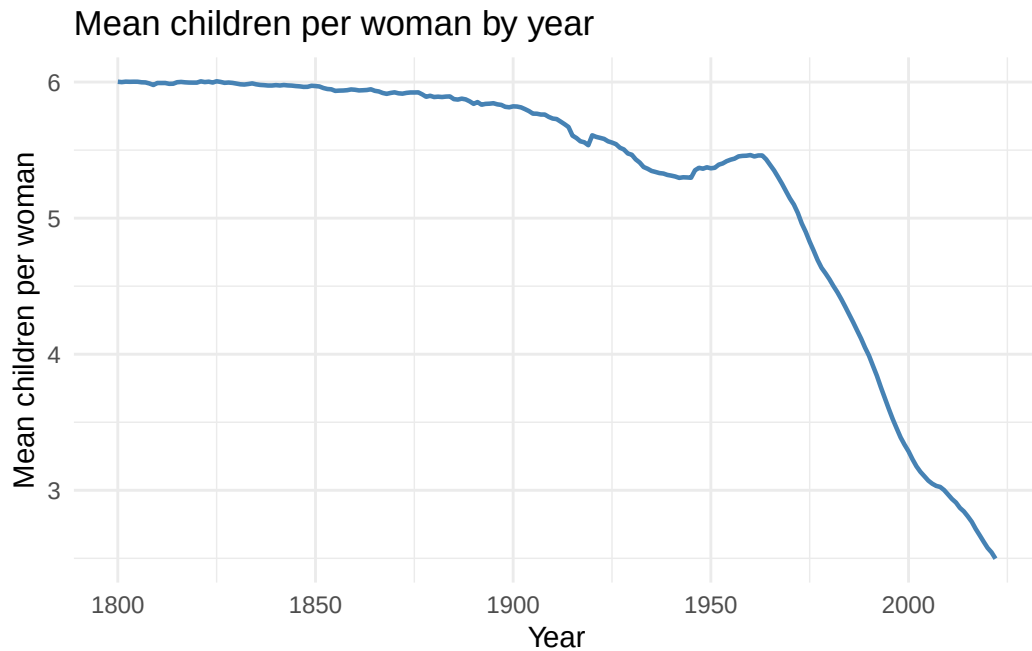


Figure 3: Mean children per woman by year

The mean number of children per woman declined substantially over time. It remained close to 6 children per woman throughout much of the 1800s, followed by a gradual decline during the first half of the twentieth century. After a small increase around the middle of the twentieth century, the average declined more rapidly, reaching about 2.5 children per woman by 2022.

```
ggplot(children_per_woman_2022, aes(children_per_woman)) +  
  geom_histogram(bins = 25, fill = "blue", color = "white") +  
  labs(  
    x = "Children per woman",  
    y = "Number of countries",  
    title = "Distribution of children per woman, 2022"  
  ) +  
  theme_minimal()
```

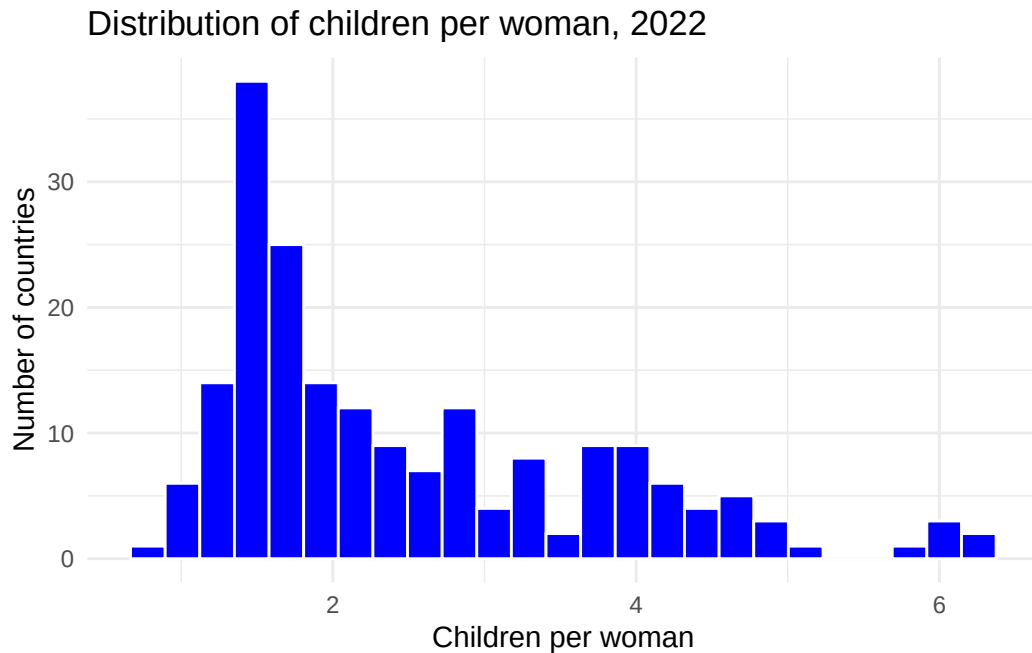


Figure 4: Distribution of children per woman, 2022

In 2022, most countries had between about 1 and 3 children per woman, with the largest concentration around 1.5 to 2 children per woman. The distribution is right-skewed because a smaller number of countries have substantially higher fertility rates, with some exceeding 6 children per woman.

```
bind_rows(
  children_per_woman_2022 |>
    slice_max(children_per_woman, n = 5) |>
    mutate(group = "Highest"),
  children_per_woman_2022 |>
    slice_min(children_per_woman, n = 5) |>
    mutate(group = "Lowest")
) |>
  select(group, name, children_per_woman)
```

```
# A tibble: 10 x 3
  group  name      children_per_woman
  <chr>  <chr>          <dbl>
1 Highest Somalia      6.26
2 Highest Chad        6.22
3 Highest Niger       6.14
```

4 Highest	Congo, Dem. Rep.	6.11
5 Highest	Central African Republic	6.02
6 Lowest	South Korea	0.78
7 Lowest	Ukraine	0.9
8 Lowest	Singapore	0.94
9 Lowest	Holy See	1
10 Lowest	China	1.03

Children per woman has changed substantially over time, declining from an average of around 6 children per woman in the 1800s to 2.50 in 2022. The 2022 distribution is right-skewed, with most countries having between about 1 and 3 children per woman and a smaller group having substantially higher fertility rates. The large variation across countries suggests that children per woman may be useful to consider as a potential predictor of population.

Relationship with Population

```
children_pop_2022 <- children_per_woman_2022 |>
  inner_join(
    population_long |> filter(year == 2022),
    by = c("geo", "name", "year")
  )
```

```
cor(
  children_pop_2022$children_per_woman,
  log10(children_pop_2022$population),
  use = "complete.obs"
)
```

```
[1] 0.1288445
```

```
ggplot(
  children_pop_2022,
  aes(population, children_per_woman)
) +
  geom_point(alpha = 0.6, color = "blue") +
  scale_x_log10(labels = scales::comma) +
  labs(
    x = "Population, log scale",
    y = "Children per woman"
  ) +
  theme_minimal()
```

Warning: Removed 1 row containing missing values or values outside the scale range (`geom_point()`).

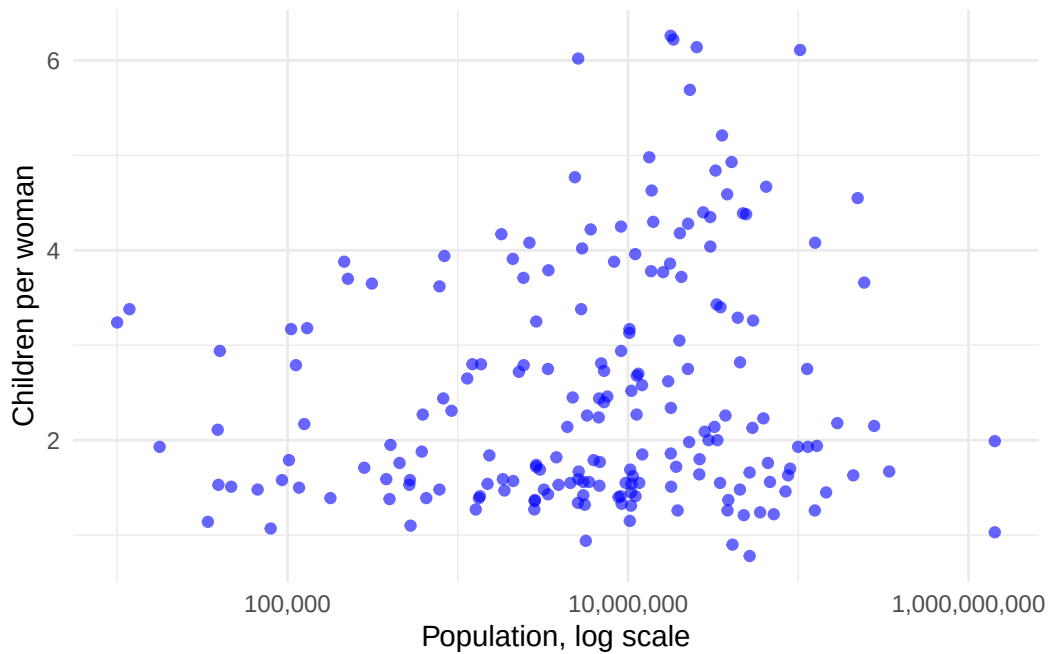


Figure 5: Children per woman against population, 2022

The correlation is 0.129, indicating a very weak positive relationship between children per woman and log population. The scatterplot shows substantial variation, suggesting that population size is not a strong predictor of fertility rate.

2.3 Happiness Score

`happiness_score` is quantitative and continuous. It represents a country's average life evaluation on a 0 to 100 scale. Values can include decimals because the variable is an average score.

```
happiness_long |>
  summarise(
    n          = n(),
    n_missing  = sum(is.na(happiness_score)),
    pct_missing = round(100 * mean(is.na(happiness_score)), 1),
    first_year = min(year[!is.na(happiness_score)]),
    last_year  = max(year[!is.na(happiness_score)])
  )
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
  <int>   <int>     <dbl>     <int>    <int>
1  2898     721      24.9      2005     2022
```

There are 721 missing happiness score observations, representing 24.9% of the data. The data are available from 2005 to 2022, and the missing values are likely due to limited coverage in earlier years and for some countries.

Because happiness score is measured on a 0–100 scale, we check for values outside that range.

```
happiness_long |>
  filter(
    !is.na(happiness_score),
    happiness_score < 0 | happiness_score > 100
  ) |>
  arrange(happiness_score)
```

```
# A tibble: 0 x 4
# i 4 variables: geo <chr>, name <chr>, year <int>, happiness_score <dbl>
```

There are no observations outside the expected 0–100 range.

We also make sure the variable is stored numerically.

```
happiness_long <- happiness_long |>
  mutate(happiness_score = as.numeric(happiness_score))
```

```
happiness_long |>
  filter(!is.na(happiness_score)) |>
  summarise(
    mean    = mean(happiness_score),
    median  = median(happiness_score),
    sd      = sd(happiness_score),
    min     = min(happiness_score),
    q1      = quantile(happiness_score, 0.25),
    q3      = quantile(happiness_score, 0.75),
    max     = max(happiness_score)
  )
```

```
# A tibble: 1 x 7
  mean median   sd   min    q1    q3   max
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  54.7   54.0  11.3  12.8  46.3   63  80.2
```

```
happiness_2022 <- happiness_long |>
  filter(!is.na(happiness_score), year == 2022)

happiness_2022 |>
  summarise(
    year      = unique(year),
    n_countries = n(),
    mean      = mean(happiness_score),
    median    = median(happiness_score),
    sd        = sd(happiness_score),
    min       = min(happiness_score),
    q1        = quantile(happiness_score, 0.25),
    q3        = quantile(happiness_score, 0.75),
    max       = max(happiness_score)
  )
```

```
# A tibble: 1 x 9
  year n_countries mean median   sd   min    q1    q3   max
<int>      <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  2022         137  54.9   57.9  12.4  12.8  46.0  63.6  77.3
```

Across all years, happiness scores had a mean of 54.68 and a median of 54.05, with values ranging from 12.81 to 80.19. The standard deviation was 11.33, showing moderate variation in happiness scores. In 2022, the mean happiness score was 54.88 and the median was 57.88. Scores ranged from 12.81 to 77.29, with the middle 50% falling between 45.96 and 63.57.

```
happiness_long |>
  filter(!is.na(happiness_score)) |>
  group_by(year) |>
  summarise(mean_happiness = mean(happiness_score)) |>
  ggplot(aes(year, mean_happiness)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  labs(
    x = "Year",
    y = "Mean happiness score",
    title = "Mean happiness score by year"
```

```
) +  
theme_minimal()
```



Figure 6: Mean happiness score by year

The mean happiness score fluctuates over time, generally staying between about 54 and 57 after 2006. The highest mean occurs in 2020, followed by a decline through 2022.

```
ggplot(happiness_2022, aes(happiness_score)) +  
  geom_histogram(bins = 25, fill = "blue", color = "white") +  
  labs(  
    x = "Happiness score",  
    y = "Number of countries",  
    title = "Distribution of happiness scores, 2022"  
  ) +  
  theme_minimal()
```

Distribution of happiness scores, 2022

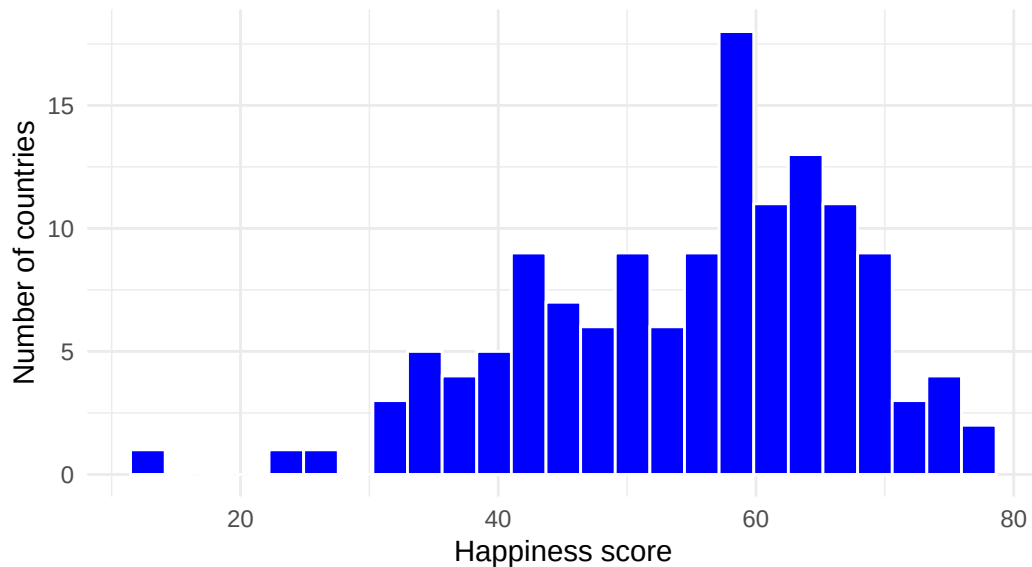


Figure 7: Distribution of happiness scores, 2022

The 2022 happiness scores are concentrated around 55–65, with most countries scoring above 50. A small number of countries have much lower scores, creating a longer left tail.

```
bind_rows(  
  happiness_2022 |>  
    slice_max(happiness_score, n = 5) |>  
    mutate(group = "Highest"),  
  happiness_2022 |>  
    slice_min(happiness_score, n = 5) |>  
    mutate(group = "Lowest")  
) |>  
select(group, name, happiness_score)
```

```
# A tibble: 10 x 3  
  group  name      happiness_score  
  <chr> <chr>          <dbl>  
1 Highest Finland      77.3  
2 Highest Israel       76.6  
3 Highest Denmark      75.4  
4 Highest Iceland      74.5  
5 Highest Sweden       74.3
```

6	Lowest	Afghanistan	12.8
7	Lowest	Lebanon	23.5
8	Lowest	Sierra Leone	25.6
9	Lowest	Lesotho	31.9
10	Lowest	Congo, Dem. Rep.	32.1

Relationship with Population

```
happiness_pop_2022 <- happiness_2022 |>
  inner_join(
    population_long |> filter(year == 2022),
    by = c("geo", "name", "year")
  )
```

```
cor(
  happiness_pop_2022$happiness_score,
  log10(happiness_pop_2022$population),
  use = "complete.obs"
)
```

```
[1] -0.1443345
```

```
ggplot(
  happiness_pop_2022,
  aes(population, happiness_score)
) +
  geom_point(alpha = 0.6, color = "blue") +
  scale_x_log10(labels = scales::comma) +
  labs(
    x = "Population, log scale",
    y = "Happiness score",
    title = "Happiness score against population, 2022"
  ) +
  theme_minimal()
```

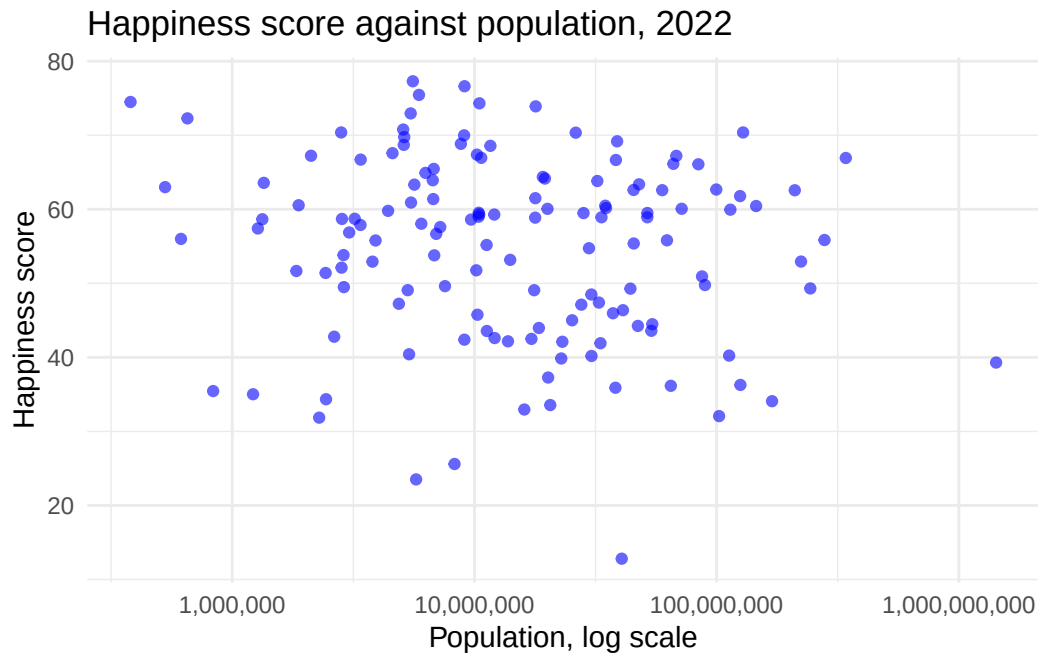


Figure 8: Happiness score against population, 2022

The correlation between happiness score and log population is -0.144, indicating a very weak negative relationship. The scatterplot shows substantial variation in happiness scores across countries with similar population sizes, suggesting that population size is not a strong predictor of happiness.

Happiness scores varied substantially across countries, with most 2022 scores between about 45 and 65. Finland had the highest score at 77.29, while Afghanistan had the lowest at 12.81. The relationship between happiness and population was very weak ($r = -0.144$), suggesting that population size is not strongly related to happiness score.

2.4 CO2 Emissions per Capita: Consumption-Based

`co2_per_capita` is quantitative and continuous, measured in tonnes of CO2 per person per year. Values run from near zero to several tens of tonnes.

```
co2_long |>
  summarise(
    n          = n(),
    n_missing  = sum(is.na(co2_per_capita)),
    pct_missing = round(100 * mean(is.na(co2_per_capita)), 1),
    first_year = min(year[!is.na(co2_per_capita)]),
```

```
last_year = max(year[!is.na(co2_per_capita)])
)
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
  <int>   <int>     <dbl>     <int>    <int>
1 42816    1805       4.2       1800     2022
```

About 4% of records are missing. Most of this is structural, since many countries are only reported from later years.

```
co2_long |>
  filter(co2_per_capita < 0) |>
  arrange(co2_per_capita)
```

```
# A tibble: 7 x 4
  geo   name   year co2_per_capita
  <chr> <chr> <int>         <dbl>
1 pan  Panama  2012        -0.579
2 pan  Panama  2005        -0.511
3 pan  Panama  2013        -0.325
4 pan  Panama  2003        -0.107
5 pan  Panama  2004        -0.085
6 pan  Panama  2006        -0.079
7 pan  Panama  2011        -0.004
```

Seven values from Panama are slightly negative. Consumption-based accounting subtracts exported emissions, so a trade hub can go net negative. These are kept as a real feature of the method, but they matter for modeling.

```
co2_latest <- co2_long |>
  filter(!is.na(co2_per_capita), year == max(year[!is.na(co2_per_capita)]))

co2_long |>
  filter(!is.na(co2_per_capita)) |>
  summarise(mean = mean(co2_per_capita), median = median(co2_per_capita),
            sd = sd(co2_per_capita), min = min(co2_per_capita),
            max = max(co2_per_capita)) |>
  knitr::kable(digits = 2, caption = "CO2 per capita, all years")
```

Table 1: CO2 per capita, all years

mean	median	sd	min	max
1.65	0.12	3.98	-0.58	62.15

```
co2_latest |>
  summarise(year = unique(year), n_countries = n(), mean = mean(co2_per_capita),
            median = median(co2_per_capita), sd = sd(co2_per_capita),
            min = min(co2_per_capita), max = max(co2_per_capita)) |>
  knitr::kable(digits = 2, caption = "CO2 per capita, 2022")
```

Table 2: CO2 per capita, 2022

year	n_countries	mean	median	sd	min	max
2022	192	4.71	2.78	5.33	0.03	27.26

```
co2_long |>
  filter(!is.na(co2_per_capita)) |>
  group_by(year) |>
  summarise(mean_co2 = mean(co2_per_capita)) |>
  ggplot(aes(year, mean_co2)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  labs(x = "Year", y = "Mean CO2 per capita, tonnes") +
  theme_minimal()
```

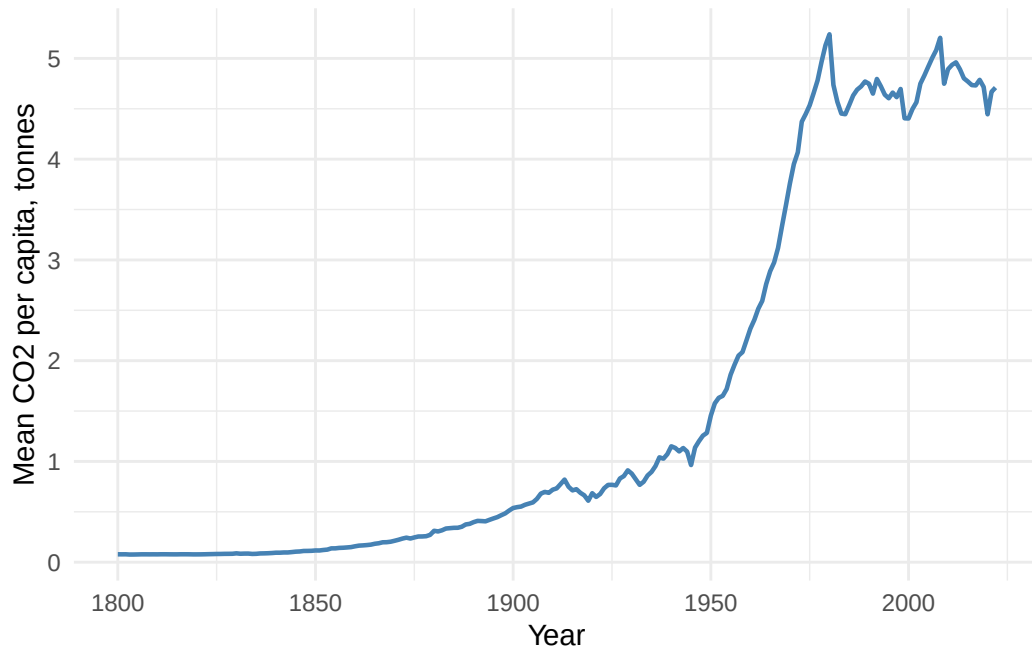


Figure 9: Mean CO2 emissions per capita by year

```
ggplot(co2_latest, aes(co2_per_capita)) +  
  geom_histogram(bins = 25, fill = "blue", color = "white") +  
  labs(x = "CO2 per capita, tonnes", y = "Number of countries") +  
  theme_minimal()
```

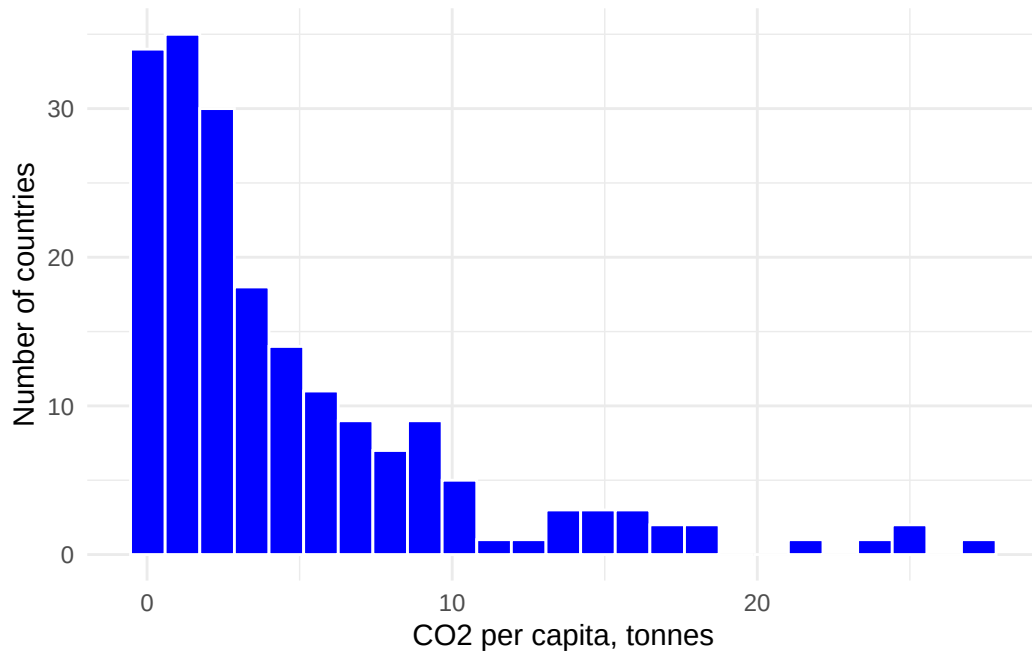


Figure 10: Distribution of CO2 emissions per capita, 2022

```
bind_rows(
  co2_latest |> slice_max(co2_per_capita, n = 5) |> mutate(group = "Highest"),
  co2_latest |> slice_min(co2_per_capita, n = 5) |> mutate(group = "Lowest")
) |>
select(group, name, co2_per_capita) |>
knitr::kable(digits = 2, caption = "Highest and lowest CO2 per capita, 2022")
```

Table 3: Highest and lowest CO2 per capita, 2022

group	name	co2_per_capita
Highest	Qatar	27.26
Highest	Singapore	25.43
Highest	Brunei	24.48
Highest	Kuwait	24.01
Highest	UAE	21.12
Lowest	Congo, Dem. Rep.	0.03
Lowest	Burundi	0.05
Lowest	Somalia	0.05
Lowest	Central African Republic	0.07
Lowest	Chad	0.07

The 2022 distribution is right-skewed, with most countries below 5 tonnes. The highest are Qatar, Singapore, Brunei, Kuwait and the UAE, small wealthy economies tied to oil, gas or trade. The lowest are the DR Congo, Burundi, Somalia, the Central African Republic and Chad. Average emissions climbed through industrialization and have leveled off since about 1980.

```
co2_pop_2022 <- co2_latest |>
  inner_join(population_long |> filter(year == 2022),
    by = c("geo", "name", "year"))

cor(co2_pop_2022$co2_per_capita, log10(co2_pop_2022$population))
```

```
[1] -0.0615751
```

```
ggplot(co2_pop_2022, aes(population, co2_per_capita)) +
  geom_point(alpha = 0.6, color = "blue") +
  scale_x_log10(labels = scales::comma) +
  labs(x = "Population, log scale", y = "CO2 per capita, tonnes") +
  theme_minimal()
```

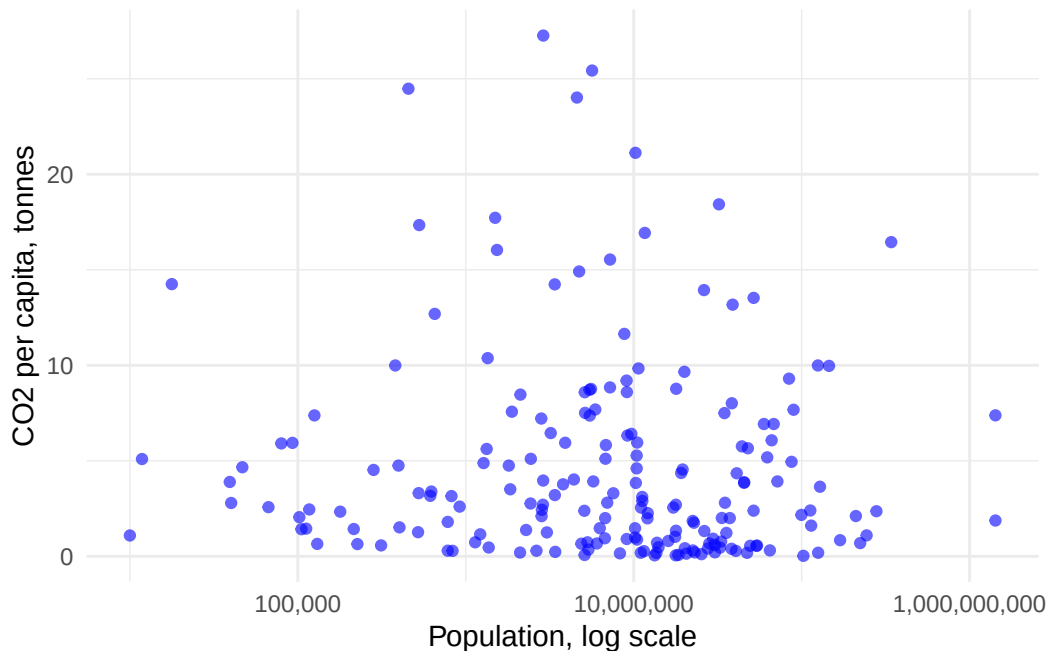


Figure 11: CO2 emissions per capita against population, 2022

There is almost no relationship between population and per-capita emissions, with $r = -0.06$ against log population. This is expected, since the measure is already per person and depends more on income and energy use than country size.

2.5 Income Groups: World Bank

`income_level` is categorical and ordinal with four levels: Low, Lower middle, Upper middle and High income, assigned by the World Bank from GNI per capita.

```
income_long <- income_long |>
  mutate(income_level = factor(
    income_level,
    levels = c("Low income", "Lower middle income",
               "Upper middle income", "High income"),
    ordered = TRUE
  ))

income_long |>
  summarise(
    n          = n(),
    n_missing  = sum(is.na(income_level)),
    pct_missing = round(100 * mean(is.na(income_level)), 1),
    first_year = min(year[!is.na(income_level)]),
    last_year  = max(year[!is.na(income_level)])
  ) |>
  knitr::kable()
```

n	n_missing	pct_missing	first_year	last_year
6984	0	0	1987	2022

There is no missing data from 1987 to 2022, the years the World Bank classifies. Earlier years will appear missing once this is joined with longer variables, which is structural.

```
income_latest <- income_long |>
  filter(!is.na(income_level), year == max(year[!is.na(income_level)]))

income_latest |>
  count(income_level) |>
  mutate(pct = round(100 * n / sum(n), 1)) |>
  knitr::kable(caption = "World Bank income group, 2022")
```

Table 4: World Bank income group, 2022

income_level	n	pct
Low income	26	13.4
Lower middle income	55	28.4
Upper middle income	53	27.3
High income	60	30.9

```
ggplot(income_latest, aes(income_level)) +
  geom_bar(fill = "orange") +
  scale_x_discrete(labels = scales::label_wrap(12)) +
  labs(x = "Income group", y = "Number of countries") +
  theme_minimal()
```

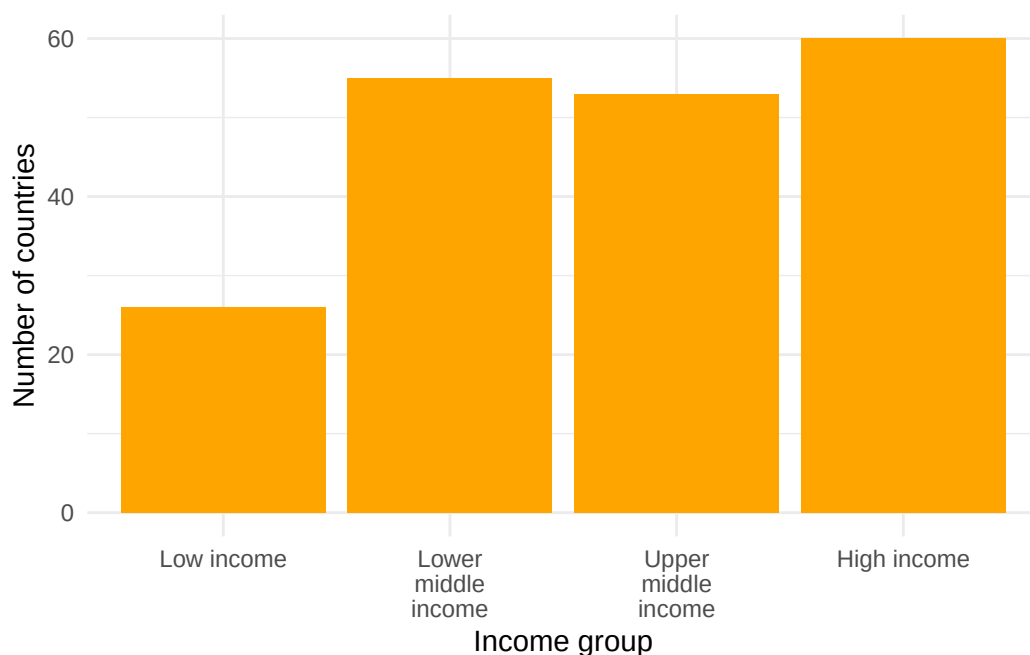


Figure 12: Number of countries by income group, 2022

```
income_long |>
  filter(!is.na(income_level)) |>
  count(year, income_level) |>
  group_by(year) |>
  mutate(pct = n / sum(n)) |>
  ggplot(aes(year, pct, fill = income_level)) +
```

```
geom_area() +
labs(x = "Year", y = "Share of countries", fill = "Income group") +
theme_minimal()
```

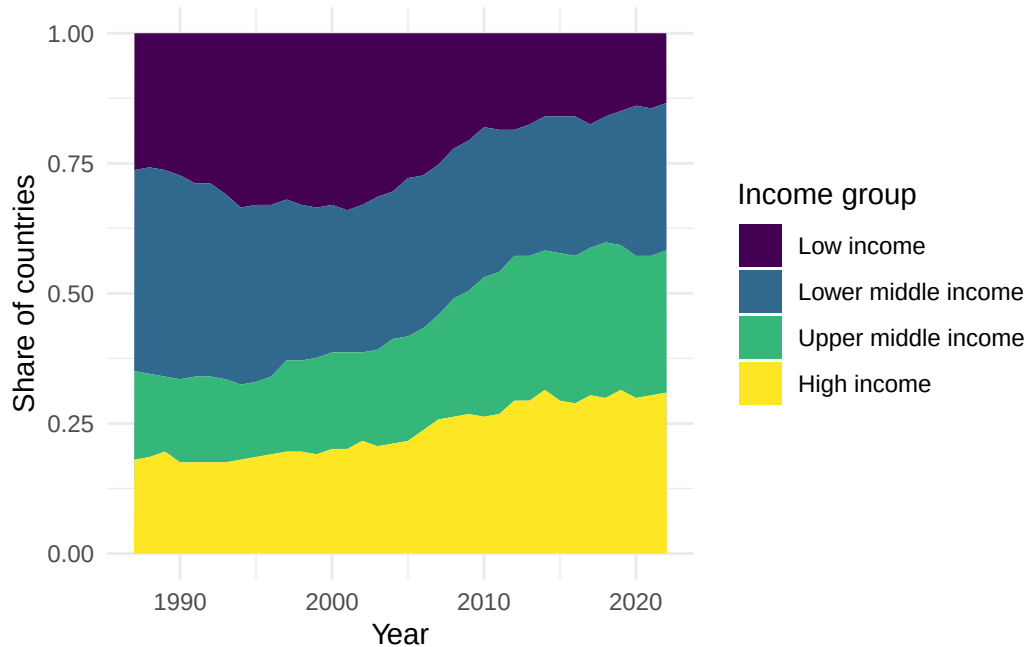


Figure 13: Share of countries in each income group, 1987 to 2022

In 2022 High income is the largest group with 60 countries, or 30.9%, followed by Lower middle at 55, Upper middle at 53 and Low income at 26. Since 1987 the High income share rose from 18% to almost 31%, while Low income fell from 26% to 13%.

```
income_pop_2022 <- income_latest |>
  inner_join(population_long |> filter(year == 2022),
    by = c("geo", "name", "year"))

income_pop_2022 |>
  group_by(income_level) |>
  summarise(n = n(), median_pop = median(population), mean_pop = mean(population)) |>
  knitr::kable(digits = 0, caption = "Population by income group, 2022")
```

Table 5: Population by income group, 2022

income_level	n	median_pop	mean_pop
Low income	26	21515450	27571761
Lower middle income	55	13759501	58989892
Upper middle income	53	5305270	52887214
High income	60	5294268	20262448

Population by income group, 2022

```
ggplot(income_pop_2022, aes(income_level, population)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6, color = "orange", size = 2) +
  stat_summary(fun = median, geom = "crossbar", width = 0.5,
               color = "black", linewidth = 0.4) +
  scale_x_discrete(labels = scales::label_wrap(12)) +
  scale_y_log10(labels = scales::comma) +
  labs(x = "Income group", y = "Population, log scale") +
  theme_minimal()
```

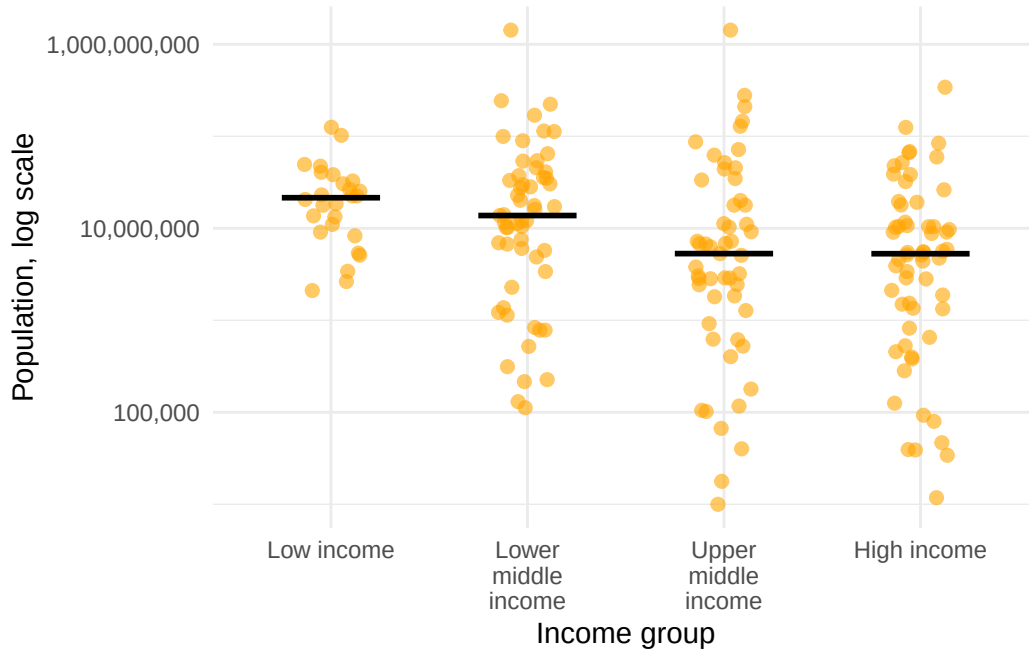


Figure 14: Population by income group, 2022

Median population is highest in Low income at 21.5 million and lowest in High income at 5.3 million, and every group mean exceeds its median. In Lower middle income the mean of 59

million is over four times the median of 13.8 million, driven by India and Pakistan. China and Indonesia do the same in Upper middle income. Population size and income group look largely unrelated.

2.6 People In Poverty

```
# Summary Statistics and Visualization
# Number of People In Poverty (Quantitative)
poverty_long |>
  summarize(
    n          = n(),
    n_missing  = sum(is.na(people_in_poverty)),
    pct_missing = round(100 * mean(is.na(people_in_poverty)), 1),
    first_year  = min(year[!is.na(people_in_poverty)]),
    last_year   = max(year[!is.na(people_in_poverty)])
  )
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
<int>   <int>      <dbl>      <int>    <int>
1 43262    1566        3.6      1800     2022
```

People in poverty is quantitative and discrete, spanning from 1800 to 2022. It represents the number of people in poverty in a country with poverty defined as living on less than \$3 at 2021 international prices. There are 1566 missing poverty observations, representing 3.6% of the data. The missing values are likely due to limited data in earlier years and for some countries.

```
poverty_long |>
  filter(
    !is.na(people_in_poverty),
    people_in_poverty < 0
  ) |>
  arrange(people_in_poverty)
```

```
# A tibble: 0 x 4
# i 4 variables: geo <chr>, name <chr>, year <int>, people_in_poverty <dbl>
```

There are no negative values for people in poverty, so no observations need to be removed by this check.

```
poverty_long |>
  filter(!is.na(people_in_poverty)) |>
  group_by(name) |>
  summarise(
    mean    = mean(people_in_poverty),
    median  = median(people_in_poverty),
    sd      = sd(people_in_poverty),
    min     = min(people_in_poverty),
    q1      = quantile(people_in_poverty, 0.25),
    q3      = quantile(people_in_poverty, 0.75),
    max     = max(people_in_poverty)
  )
```

A tibble: 194 x 8

	name <chr>	mean <dbl>	median <dbl>	sd <dbl>	min <dbl>	q1 <dbl>	q3 <dbl>	max <dbl>
1	Afghanistan	6749968.	5258906	3979899.	3.08e6	3.53e6	9.03e6	2.04e7
2	Albania	465075.	446436	259770.	7.08e3	2.83e5	6.87e5	8.71e5
3	Algeria	3027239.	2914254	1255232.	1.73e3	2.52e6	3.90e6	5.93e6
4	Andorra	5.27	5	3.43	1 e0	3.25e0	6 e0	1.3 e1
5	Angola	2706724.	2275585	2191297.	7.87e5	1.61e6	2.71e6	1.52e7
6	Antigua and Barbuda	22510.	25818	12120.	1.51e3	1.20e4	3.44e4	3.53e4
7	Argentina	517627.	372961	614034.	1.02e4	2.68e5	5.72e5	6.30e6
8	Armenia	434796.	453368	207121.	5.02e4	2.87e5	5.76e5	1.08e6
9	Australia	167228.	155217	56575.	6.90e4	1.21e5	2.06e5	3.19e5
10	Austria	773737.	528039	712695.	3.54e2	5.67e4	1.37e6	2.10e6

i 184 more rows

```
poverty_2022 <- poverty_long |>
  filter(!is.na(people_in_poverty), year == 2022)

poverty_2022 |>
  summarise(
    year      = unique(year),
    n_countries = n(),
    mean      = mean(people_in_poverty),
    median    = median(people_in_poverty),
    sd        = sd(people_in_poverty),
    min       = min(people_in_poverty),
    q1        = quantile(people_in_poverty, 0.25),
    q3        = quantile(people_in_poverty, 0.75),
```

```
max = max(people_in_poverty)
)
```

A tibble: 1 x 9

	year	n_countries	mean	median	sd	min	q1	q3	max
	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022	186	4699704.	203594.	13332120.	6 9885.	2362486.	100305610	

In 2022, the poverty data set only contained 186 countries with the mean people in poverty over all the countries being 4,699,704 people. Meanwhile, the median is only 203,593.5, which would imply that the graph is right skewed especially since the third quartile is also lower than the mean.

```
combined |>
  filter(!is.na(people_in_poverty), year == 2022) |>
  ggplot(mapping = aes(x = people_in_poverty, y = population)) +
  geom_point(shape = 5) +
  labs(title = "People in Poverty vs Population",
       x = "People in Poverty", y = "Population") +
  theme_minimal()
```

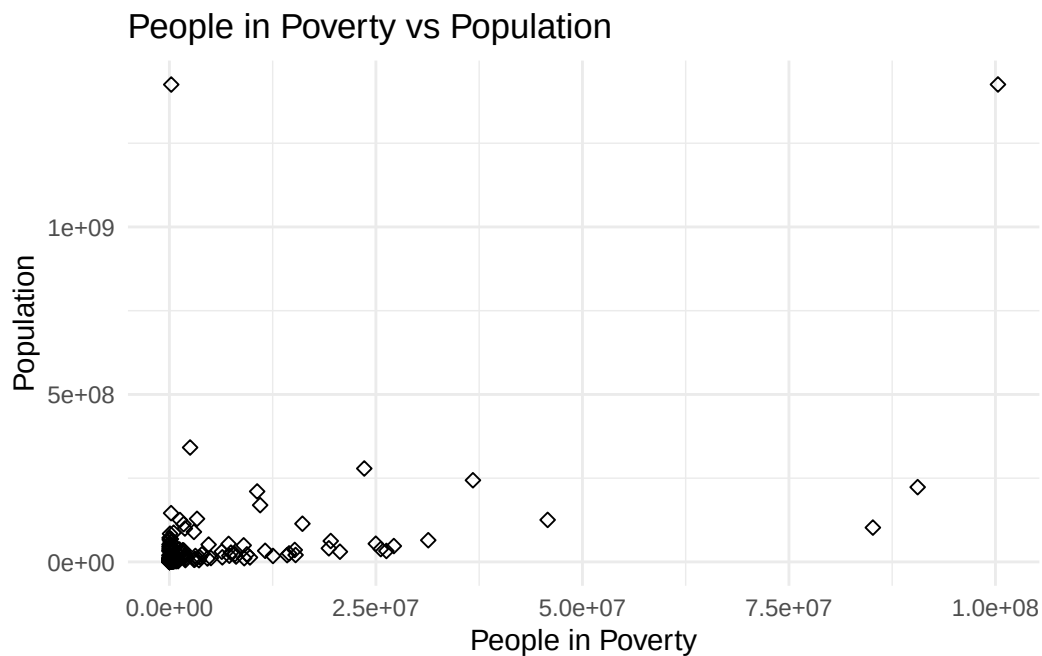


Figure 15: People in Poverty 2022

This confirms our suspicions that in 2022, the graph would be right skewed. We can see that most of the points lie on the left side. When looking into the data, most of the points on the right side are from developing countries such as India and Nigeria as the top 2 countries.

2.7 At Least Basic Sanitation Percentage Overall

```
#Basic Sanitation Overall (Quantitative)
sanitation_long |>
  summarize(
    n          = n(),
    n_missing  = sum(is.na(sanitation_access_pct)),
    pct_missing = round(100 * mean(is.na(sanitation_access_pct)), 1),
    first_year  = min(year[!is.na(sanitation_access_pct)]),
    last_year   = max(year[!is.na(sanitation_access_pct)])
  )
```

```
# A tibble: 1 x 5
      n n_missing pct_missing first_year last_year
<int>   <int>      <dbl>      <int>    <int>
1  4439      84        1.9        2000      2022
```

Sanitation Access Percentage is quantitative and continuous, spanning from 2000 to 2022. It represents the percentage of people using improved sanitation facilities that are not shared with other households. Improved sanitation facilities are systems designed to separate human waste from human contact. There are 84 missing observations, representing 1.9% of the data. The missing values are likely due to limited data in earlier years and no reporting for some countries. A notable country in the analysis is Andorra with basic sanitation access at 100% between 2000 and 2022.

```
sanitation_long |>
  filter(
    !is.na(sanitation_access_pct),
    sanitation_access_pct < 0 | sanitation_access_pct > 100
  ) |>
  arrange(sanitation_access_pct)
```

```
# A tibble: 0 x 4
# i 4 variables: geo <chr>, name <chr>, year <int>, sanitation_access_pct <dbl>
```

There are no values shown to be outside of the expected 0 to 100 range.

```
sanitation_long |>
  filter(!is.na(sanitation_access_pct)) |>
  group_by(name) |>
  summarise(
    mean    = mean(sanitation_access_pct),
    median  = median(sanitation_access_pct),
    sd      = sd(sanitation_access_pct),
    min     = min(sanitation_access_pct),
    q1      = quantile(sanitation_access_pct, 0.25),
    q3      = quantile(sanitation_access_pct, 0.75),
    max     = max(sanitation_access_pct)
  )
```

A tibble: 193 x 8

	name	mean	median	sd	min	q1	q3	max
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Afghanistan	36.4	36.5	9.38	22.3	28.8	44.1	51.7
2	Albania	95.3	95.7	3.31	89.5	92.6	98.3	99.3
3	Algeria	86.2	86.3	0.755	84.5	85.8	86.8	87.2
4	Andorra	100	100	0	100	100	100	100
5	Angola	42.8	43.2	5.61	33.8	38.3	48.0	50.3
6	Antigua and Barbuda	93.0	93.3	4.93	84.8	89.0	97.4	99.6
7	Argentina	91.3	91.4	2.28	87.7	89.5	93.2	94.8
8	Armenia	90.6	90.5	1.62	88.0	89.3	91.9	93.3
9	Australia	100.0	100.0	0.00295	100.0	100.0	100.0	100
10	Austria	100.0	100.0	0.0104	100.0	100.0	100.0	100

i 183 more rows

```
sanitation_2022 <- sanitation_long |>
  filter(!is.na(sanitation_access_pct), year == 2022)

sanitation_2022 |>
  summarise(
    year      = unique(year),
    n_countries = n(),
    mean      = mean(sanitation_access_pct),
    median    = median(sanitation_access_pct),
    sd        = sd(sanitation_access_pct),
    min       = min(sanitation_access_pct),
    q1        = quantile(sanitation_access_pct, 0.25),
```

```

    q3      = quantile(sanitation_access_pct, 0.75),
    max      = max(sanitation_access_pct)
  )

```

A tibble: 1 x 9

	year	n_countries	mean	median	sd	min	q1	q3	max
	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022	182	78.6	91.8	26.6	9.68	63.4	98.8	100

In 2022, the sanitation data set only contained 182 countries with the mean access to at least basic sanitation over all the countries being 78.60%. Meanwhile, the median is 91.78%, which would imply that the graph is left skewed.

```

combined |>
  filter(!is.na(sanitation_access_pct), year == 2022) |>
  ggplot(mapping = aes(x = sanitation_access_pct, y = population)) +
  geom_point(shape = 5) +
  labs(title = "Access to at Least Basic Sanitation vs Population",
       x = "At Least Basic Sanitation (%)", y = "Population") +
  theme_minimal()

```

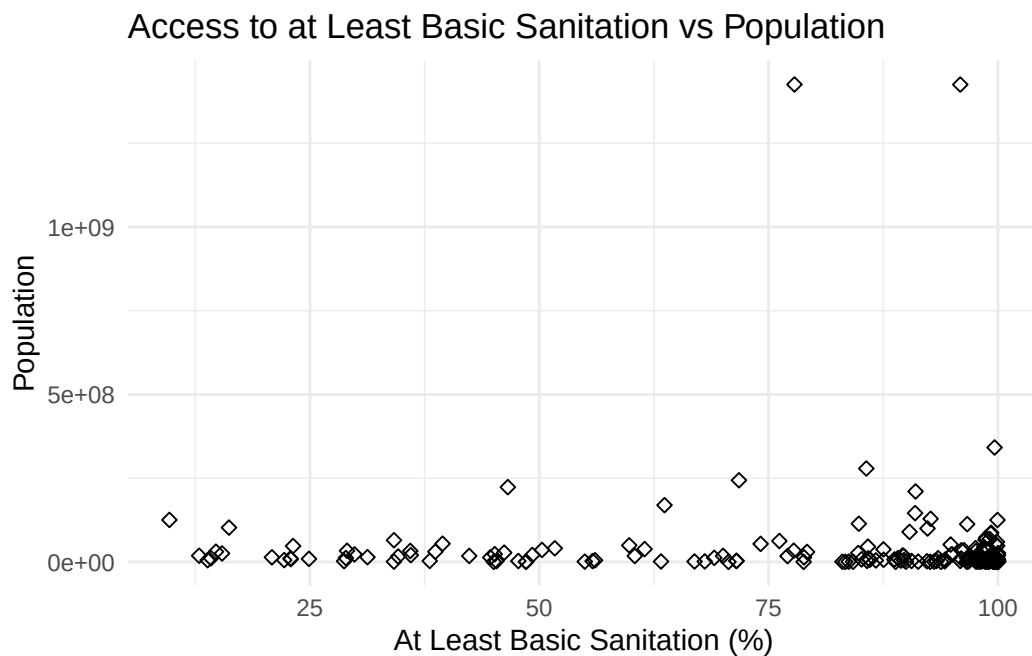


Figure 16: Access to at Least Basic Sanitation Percentage 2022

This confirms our suspicions that in 2022, the graph would be left skewed. We can see that a clustering of points between 75 and 100. When looking into the data, most of the points on the left side are from developing countries such as Ethiopia and Chad as the bottom 2 countries.

2.8 EDA Discussion

```
correlations <- tibble(
  variable = c(
    "Children per woman",
    "Happiness score",
    "CO2 emissions per capita",
    "People in poverty",
    "Basic sanitation access"
  ),
  correlation = c(
    cor(children_pop_2022$children_per_woman,
        log10(children_pop_2022$population),
        use = "complete.obs"),

    cor(happiness_pop_2022$happiness_score,
        log10(happiness_pop_2022$population),
        use = "complete.obs"),

    cor(co2_pop_2022$co2_per_capita,
        log10(co2_pop_2022$population),
        use = "complete.obs"),

    cor(
      inner_join(
        poverty_long |> filter(year == 2022),
        population_long |> filter(year == 2022),
        by = c("geo", "name", "year")
      )$people_in_poverty,
      log10(
        inner_join(
          poverty_long |> filter(year == 2022),
          population_long |> filter(year == 2022),
          by = c("geo", "name", "year")
        )$population
      ),
    ),
    use = "complete.obs"
```

```

),

cor(
  inner_join(
    sanitation_long |> filter(year == 2022),
    population_long |> filter(year == 2022),
    by = c("geo", "name", "year")
  )$sanitation_access_pct,
  log10(
    inner_join(
      sanitation_long |> filter(year == 2022),
      population_long |> filter(year == 2022),
      by = c("geo", "name", "year")
    )$population
  ),
  use = "complete.obs"
)
) |>
mutate(correlation = round(correlation, 3))

knitr::kable(
  correlations,
  caption = "Correlations with log population, 2022"
)

```

Table 6: Correlations with log population, 2022

variable	correlation
Children per woman	0.129
Happiness score	-0.144
CO2 emissions per capita	-0.062
People in poverty	0.407
Basic sanitation access	-0.146

Based on the exploratory analysis and correlations, we selected people in poverty as the predictor for the simple linear regression. For the multiple linear regression models, we added children per woman and basic sanitation access separately to examine whether these variables provide additional explanatory power beyond poverty. These three variables had the most correlation with our response variable of total population.

3. Regression Models

3.1 SLR - Predictor: Poverty

```
slr_data <- combined |>
  filter(
    year == 2022,
    !is.na(population),
    !is.na(people_in_poverty) )

slr <- lm(
  population ~ people_in_poverty,
  data = slr_data)

summary(slr)
```

Call:

```
lm(formula = population ~ people_in_poverty, data = slr_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-378132906	-17225912	-14659798	-5712754	1406773555

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.720e+07	1.049e+07	1.639	0.103
people_in_poverty	5.441e+00	7.439e-01	7.314	7.74e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

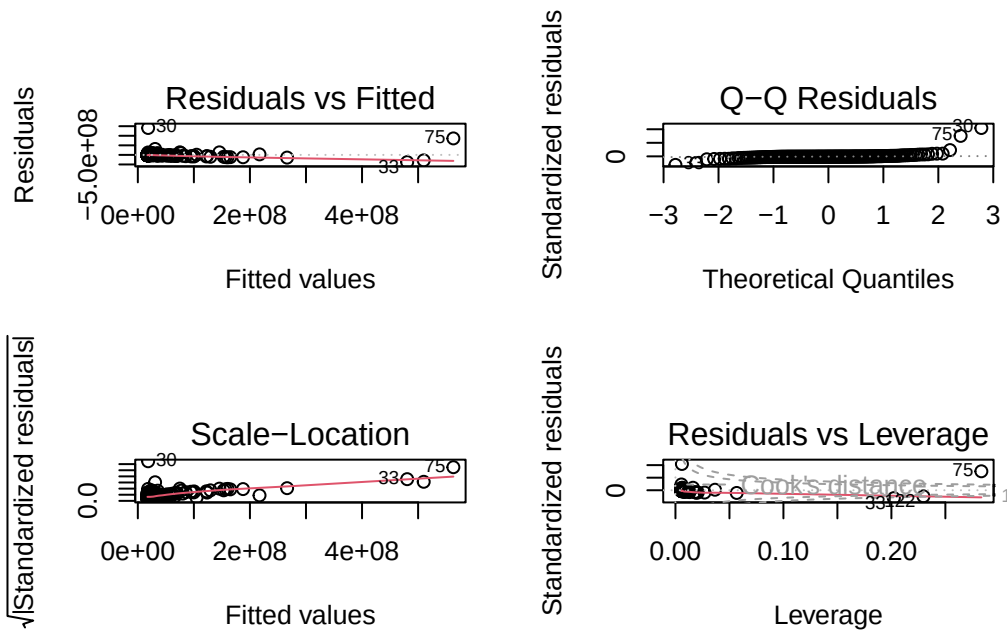
Residual standard error: 134900000 on 184 degrees of freedom

Multiple R-squared: 0.2252, Adjusted R-squared: 0.221

F-statistic: 53.49 on 1 and 184 DF, p-value: 7.741e-12

The estimated slope for people in poverty is 5.441. This means that for each additional person in poverty, the predicted population increases by approximately 5.441 people, on average.

```
par(mfrow = c(2,2))
plot(slr)
```



Linearity: The Residuals vs. Fitted plot shows a noticeable curved pattern, so the linearity assumption is not well satisfied.

Independence: Each observation represents a different country, so the observations are treated as independent.

Normality: The Q-Q plot shows substantial departures from the reference line, particularly in the tails, so the residuals are not approximately normal.

Equal variance: The Scale-Location plot shows increasing residual spread as fitted population increases, indicating that the equal variance assumption is not satisfied.

3.2 MLR - Predictor: Poverty + Children per Woman

```
mlr1_data <- combined |>
  filter(
    year == 2022,
    !is.na(population),
    !is.na(people_in_poverty),
    !is.na(children_per_woman)
  )

mlr1 <- lm(
  population ~ people_in_poverty + children_per_woman,
```

```
data = mlr1_data
)

summary(mlr1)
```

Call:

```
lm(formula = population ~ people_in_poverty + children_per_woman,
    data = mlr1_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-355273151	-42594526	-14265010	24984701	1355768872

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.067e+08	2.180e+07	4.895	2.15e-06 ***
people_in_poverty	6.824e+00	7.670e-01	8.898	5.49e-16 ***
children_per_woman	-3.768e+07	8.165e+06	-4.615	7.37e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

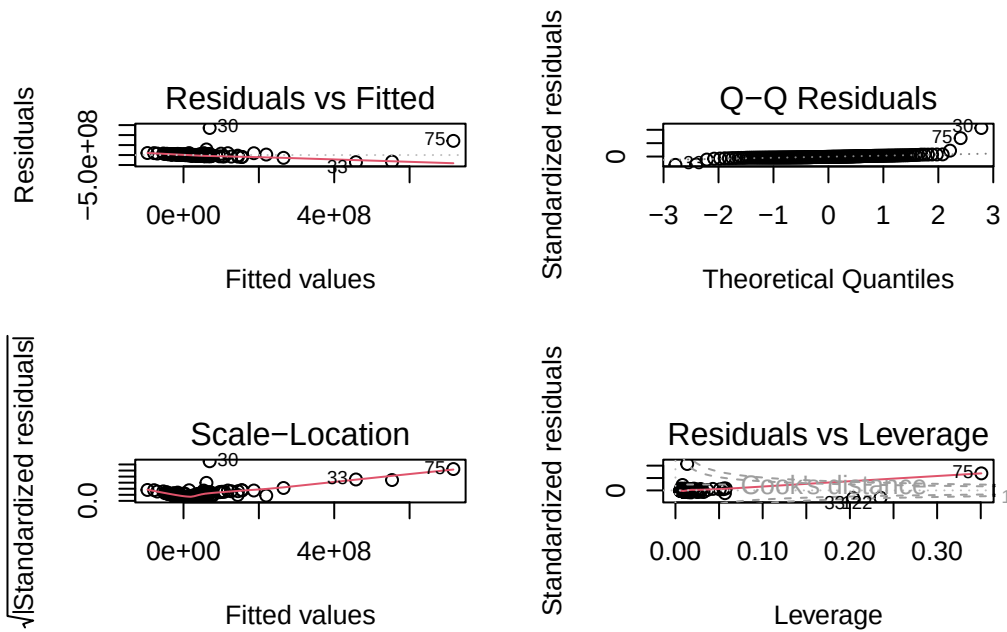
Residual standard error: 1.28e+08 on 183 degrees of freedom

Multiple R-squared: 0.306, Adjusted R-squared: 0.2984

F-statistic: 40.35 on 2 and 183 DF, p-value: 3.049e-15

Holding children per woman constant, each additional person in poverty is associated with an estimated 6.824 person increase in population. Holding people in poverty constant, each additional child per woman is associated with an estimated 37.68 million person decrease in population.

```
par(mfrow = c(2,2))
plot(mlr1)
```



Linearity: The residuals show a noticeable pattern rather than random scatter around zero, so the linearity assumption is not satisfied.

Independence: Each observation represents a different country, so the observations are treated as independent.

Normality: The Q-Q plot shows substantial departures from the reference line, particularly in the tails, so the residuals are not approximately normal.

Equal Variance: The Scale-Location plot shows increasing residual spread as fitted population increases, so the equal variance assumption is not satisfied.

3.3 MLR - Predictor: Poverty + Sanitation

```
m1r2_data <- combined |>
  filter(
    year == 2022,
    !is.na(population),
    !is.na(people_in_poverty),
    !is.na(sanitation_access_pct)
  )

m1r2 <- lm(
  population ~ people_in_poverty + sanitation_access_pct,
```

```
data = mlr2_data
)

summary(mlr2)
```

Call:

```
lm(formula = population ~ people_in_poverty + sanitation_access_pct,
    data = mlr2_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-379768521	-38033389	-21834105	14052831	1383519753

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.073e+08	3.566e+07	-3.008	0.003030 **
people_in_poverty	6.629e+00	8.138e-01	8.146	7.57e-14 ***
sanitation_access_pct	1.538e+06	4.162e+05	3.695	0.000296 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

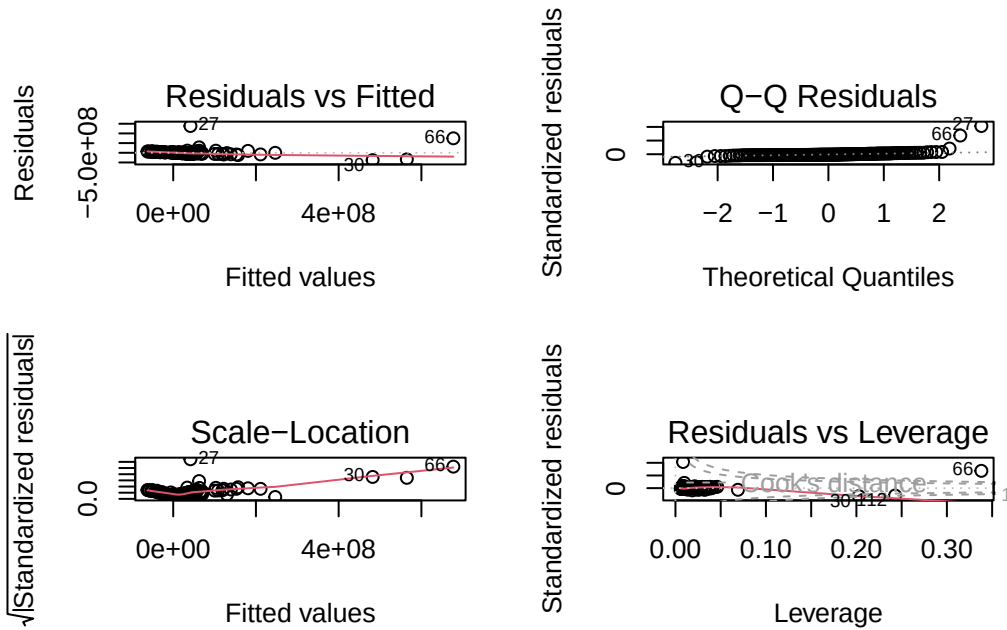
Residual standard error: 134600000 on 171 degrees of freedom

Multiple R-squared: 0.2802, Adjusted R-squared: 0.2718

F-statistic: 33.28 on 2 and 171 DF, p-value: 6.189e-13

Holding basic sanitation access constant, each additional person in poverty is associated with an estimated 6.629 person increase in population. Holding people in poverty constant, each one-percentage-point increase in basic sanitation access is associated with an estimated 1.538 million person increase in population.

```
par(mfrow = c(2,2))
plot(mlr2)
```



Linearity: The residuals show a noticeable pattern rather than random scatter, so the linearity assumption is not satisfied.

Independence: Each observation represents a different country, so the observations are treated as independent.

Normality: The Q-Q plot shows substantial departures from the reference line, so the residuals are not approximately normal.

Equal Variance: The residual spread increases as fitted population increases, so the equal variance assumption is not satisfied.

3.4 Regression Discussion

We compare the models using R^2 because it measures the proportion of variation in population that is explained by the predictors in each model. A higher R^2 indicates that the model explains more of the variation in the response variable. Our multivariate regression using Children per Woman and Poverty as predictors has the highest R^2 (0.306), compared with 0.280 for our other multivariate model using Poverty and Sanitation as predictors, and 0.225 for the simple linear regression model using only Poverty. This means that our model using Children per Woman and Poverty as predictors was the best performing model.

4. Final Question

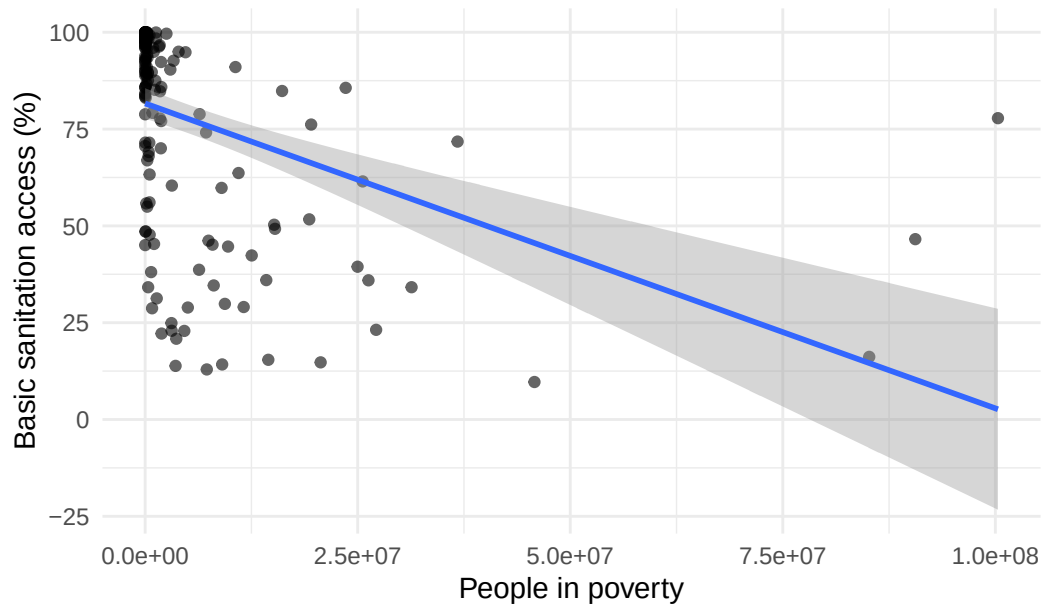
Is the number of people living in poverty associated with access to basic sanitation across countries in 2022?

```
poverty_sanitation_2022 <- poverty_long |>
  filter(year == 2022) |>
  inner_join(
    sanitation_long |> filter(year == 2022),
    by = c("geo", "name", "year")
  ) |>
  drop_na(people_in_poverty, sanitation_access_pct)

ggplot(
  poverty_sanitation_2022,
  aes(x = people_in_poverty, y = sanitation_access_pct)
) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "People in poverty",
    y = "Basic sanitation access (%)",
    title = "People in Poverty vs. Basic Sanitation Access, 2022"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

People in Poverty vs. Basic Sanitation Access, 2022



```
cor(  
  poverty_sanitation_2022$people_in_poverty,  
  poverty_sanitation_2022$sanitation_access_pct,  
  use = "complete.obs"  
)
```

```
[1] -0.4027051
```

The scatterplot shows a negative relationship between the number of people living in poverty and basic sanitation access. The correlation is -0.403, indicating a moderate negative association: countries with more people in poverty tend to have lower access to basic sanitation. This suggests that poverty and sanitation access are related across countries in 2022, although the relationship does not imply causation.